Probing Black Hole Entropy through Entanglement Entropy Scaling

Tahami Shoaib (MS21114)

Co-supervisor:
Dr. Kinjalk Lochan,
IISER Mohali

Co-supervisor:
Dr. S. Shankara-narayanan,
IIT Bombay

Abstract

In this thesis, we investigate the possibility of establishing a correspondence between quantum entanglement and black hole thermodynamics, with the aim of understanding whether entanglement can provide a microscopic explanation for black hole entropy. To address this, we develop a numerical framework for computing the entanglement entropy of quantum scalar fields in axisymmetric rotating black hole spacetimes. We first consider a slowly rotating black hole metric and discretize the spatial geometry on a lattice, thereby mapping the quantum field system onto an equivalent system of two-dimensional coupled harmonic oscillators. This framework is then extended to the full Kerr spacetime. Finally, we examine the scaling behavior of the resulting entanglement entropy with the area of the event horizon and assess its consistency with the expected area law of black hole entropy.

Black Hole Entropy

In 1972 J. Bekenstein conjectured that black holes should possess entropy. He showed that this entropy should be proportional to the area of the event horizon of the black hole.

$$S_{BH} = \frac{A}{4G\hbar}$$

However, the microscopic origin of the black hole entropy is still a mystery. Over the years, various explanations of its origin have been proposed. One such explanation involves the existence of quantum entanglement between the degrees of freedom of a quantum field with the event horizon acting as a partitioning surface.

What is entanglement entropy?

A state is called "entangled" if it cannot be written as a single tensor product of its individual subsystem states.

$$|\psi\rangle \neq |\psi^A\rangle \otimes |\psi^B\rangle$$

Entanglement entropy is a measure of entanglement within a system. It is calculated using the von Neumann entropy of a subsystem, whose reduced density matrix is obtained by tracing out the other subsystem. It turns out that entanglement entropy, just like BH entropy, scales linearly with area of the entangling surface.

$$S_{vN} = -Tr[\rho_A ln \rho_A]$$

where

$$\rho_A = Tr_B(\rho)$$

Entanglement entropy of quantum fields

Short-distance divergences are responsible for the divergence of entanglement entropy in quantum field theory. To make progress, we employ a numerical approach based on discretization, where the field is placed on a spatial lattice inside a finite region. This converts the continuum field into a quantum mechanical system of coupled oscillators. The Hamiltonian of an N-oscillator harmonic chain is:

$$H = \sum_{i=1}^{N} \frac{p_i^2}{2m} + \sum_{i=1}^{N}\sum_{j=1}^{N} \frac{1}{2} x_i K_{ij} x_j$$

where K is a real, symmetric coupling matrix encoding all interactions between the oscillators. This matrix is the central input in computing the entanglement entropy. After defining a partition, the entanglement entropy can be computed.

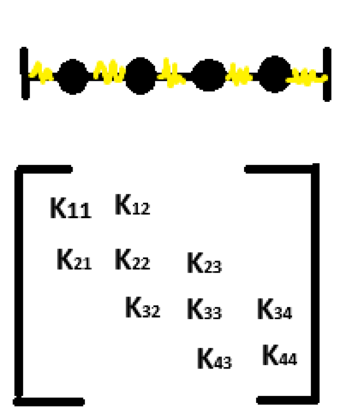

Figure 1. Structure of the coupling matrix

Entanglement entropy scaling in black hole spacetimes

The numerical approach is used to verify entropy scaling in black hole spacetimes. We start with the Schwarzschild metric, which is the solution for a static, stationary black hole. The Schwarzschild line element is:

$$ds^2 = -(1 - \frac{2M}{r})dt^2 + d\rho^2 + r^2 d\theta^2 + r^2 \sin^2\theta d\phi^2$$

Using the spherical symmetry of the spacetime, the field can be decomposed into purely radial and angular parts, using which an effective 1-D Hamiltonian is obtained:

$$H = \frac{1}{2}\sum_{lm} \int d\rho \Big(\dot{\psi}_{lm}^2(\rho) + r^2\sqrt{f}\Big(\frac{\partial}{\partial\rho}(\frac{\psi_{lm}(r)(f)^{\frac{1}{4}}}{r})\Big)^2 + \frac{f l(l+1)\psi_{lm}^2(\rho)}{r^2}\Big)$$

Lattice spacing is introduced, and derivatives are replaced by finite differences. The discretized Hamiltonian is:

$$H = \frac{1}{2a}\sum_{lm}\sum_{j}\Big[\pi_{lm,j}^2 + r_{j+\frac{1}{2}}^2 \sqrt{f_{j+\frac{1}{2}}}\Big(f_j^{\frac{1}{4}}\frac{\psi_{lm,j}}{r_j} - f_{j+1}^{\frac{1}{4}}\frac{\psi_{lm,j+1}}{r_{j+1}}\Big)^2 + f_j \frac{l(l+1)\psi_{lm,j}^2}{r_j^2}\Big]$$

Interpreting the variables $\psi_{lm,j}$ as discrete oscillator degrees of freedom, the coupling matrix can now be constructed.

Spinning black holes

The Kerr metric is the solution for rotating black holes and possesses axial symmetry.

$$ds^2 = -\left(1 - \frac{2Mr}{\Sigma}\right)dt^2 - \frac{4Mar}{\Sigma}dt\,d\phi + \frac{\Sigma}{\Delta}dr^2 + \Sigma\,d\theta^2 + \frac{A}{\Sigma}sin^2\theta d\phi^2$$

where $\Sigma = r^2 + a^2\cos^2\theta,\ \ \Delta = r^2 - 2Mr + a^2$
In addition to the event horizon, the Kerr black hole consists of a region called the ergoregion, where every timelike trajectory must move in the ϕ direction.

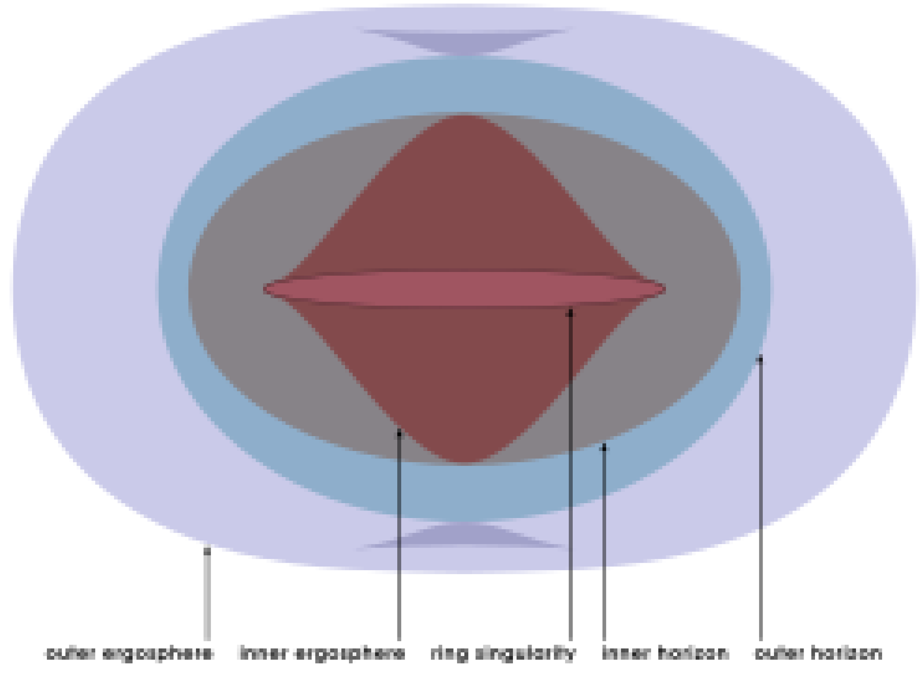

Figure 2. Structure of the Kerr black hole

Discretization in axisymmetric spacetimes

The discretization procedure is extended to analyze rotating black holes. This time the discretization is performed in two dimensions since θ direction no longer has symmetry. Moreover, the erogregion leads to negative energy modes. Hence, we switch to a frame that co-rotates with the black hole to verify entropy scaling.

$$H = \sum_{mjk}\frac{1}{2ab}\Big(a^2b^2\pi_m'^2 + b^2h^2\sqrt{\Delta_{j+.5}}\Big(\psi'_{m,jk}\sqrt{\frac{\sqrt{\Delta_j}}{A_{jk}}} - \psi'_{m,j+1k}\sqrt{\frac{\sqrt{\Delta_{j+1}}}{A_{j+1k}}}\Big)^2 + sin\theta_{k+.5}\Delta_j\Big(\frac{\psi'_{mjk}}{\sqrt{A_{jk}sin\theta_k}} - \frac{\psi'_{mjk+1}}{\sqrt{A_{jk+1}sin\theta_{k+1}}}\Big)^2 + \frac{b^2m^2\Sigma_{jk}^2\Delta_j}{A_{jk}^2 sin^2\theta_k}\psi'^2_{mjk}\Big)$$

After extracting the coupling matrix, the entanglement entropy is computed for different values of horizon radius.

Results

In the ZAMO frame, the discretization scheme is implemented and it is found that the entanglement entropy indeed scales linearly with the horizon area for all values of the rotation parameter.

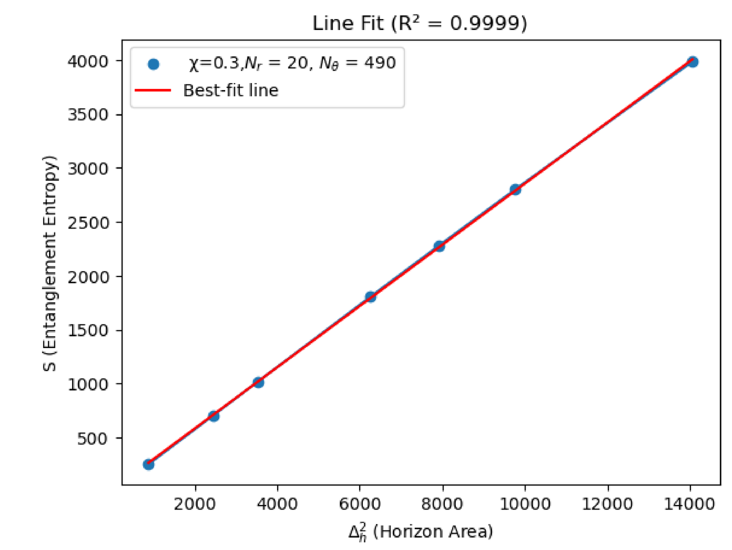

Figure 3. Entropy vs. horizon area. χ=0.3

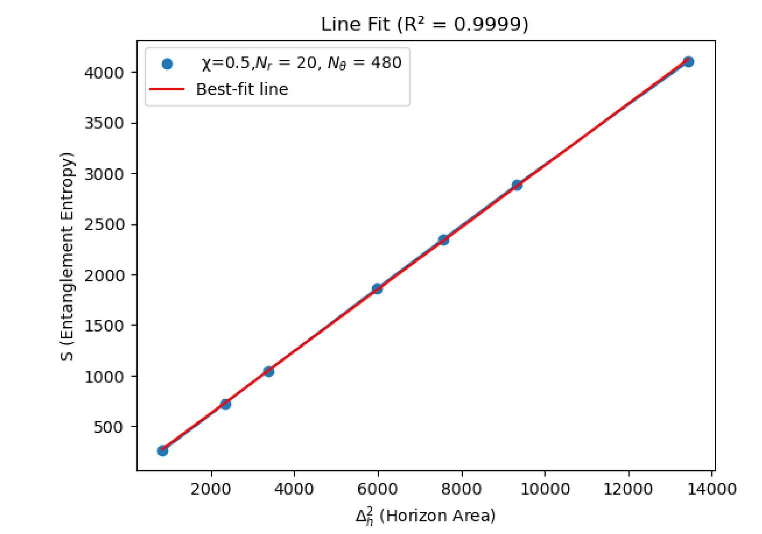

Figure 4. Entropy vs. horizon area. χ=0.5

We also intend to study the entanglement energy scaling for Kerr spacetime as well as obtain an analogous entanglement temperature. Entanglement energy calculation mainly requires the coupling matrix, which is already at our disposal. We also intend to extend this analysis to other axisymmetric spacetimes.

References

1. Srednicki, M. (1993). Entropy and area. Physical Review Letters, 71(5), 666–669. https://doi.org/10.1103/physrevlett.71.666
2. Chandran, S. M., Shankaranarayanan, S. (2020). One-to-one correspondence between entanglement mechanics and black hole thermodynamics. Physical Review. D/Physical Review. D., 102(12). https://doi.org/10.1103/physrevd.102.125025
3. Das, S., Shankaranarayanan, S. (2007). Where are the black-hole entropy degrees of freedom? Classical and Quantum Gravity, 24(20), 5299–5306. https://doi.org/10.1088/0264-9381/24/20/022